Curriculum Vitae

Christian Margreiter, MSc

PhD Candidate in Biology

Date of birth	11 January 1995
Nationality	Austrian
E-mail	christian.margreiter@mci.edu
	christian.margreiter@student.uibk.ac.at
ORCID	0009-0009-1700-3156



Education

since 2024 — **PhD Programme in Biology**
University of Innsbruck, Department of Microbiology, Austria
Cumulative thesis on bio-based gasification chars in wastewater treatment and anaerobic digestion. Supervisors: Priv.-Doz. Mag. Dr. Andreas O. Wagner; FH-Prof. Dr. techn. Angela Hofmann

2021 – 2023 — **MSc Microbiology**
University of Innsbruck, Austria
Thesis: *Effect of gasification residues and activated carbon on biogas production in anaerobic digestion systems*
Supervisors: Priv.-Doz. Mag. Dr. Andreas O. Wagner, Maraike Probst, PhD, FH-Prof. Dr. techn. Angela Hofmann. In cooperation with the Josef Ressel Centre for the Production of Activated Carbon from Municipal Residues, MCI Innsbruck

2017 – 2021 — **BSc Biology**
University of Innsbruck, Austria
Thesis: *Chlamydomonas nivalis and description of its microhabitat in the Tyrolean Alps*
Supervisor: Ao. Univ.-Prof. Mag. Dr. Birgit Sattler, Department of Ecology

2014 – 2017 — **Matura (university entrance qualification)**
Federal Secondary School of Economics for Working People, Innsbruck, Austria

2009 – 2014 — **Training as Nursery School Teacher**
Federal Training Centre for Elementary Education, Innsbruck, Austria

Professional Experience

since 01/2022 — **Research Associate (Project Employee)**
MCI | The Entrepreneurial School, Department of Environmental, Process & Energy Engineering, Innsbruck, Austria
Research on activated carbon and gasification chars in wastewater treatment and anaerobic digestion; laboratory analytics (HPLC, GC), method development; teaching and thesis supervision (see below).

07/2021 – 08/2021 **Internship, Analytics**
Novartis, Schaftenau, Austria

08/2014 – 11/2021 **Bike Messenger / Dispatcher**
DDE Logistik GmbH / DEX – Document Express e.U., Innsbruck, Austria

Teaching & Supervision

Lecturer

- *Chemical Laboratory – Instrumental Analysis / Environmental Analysis*, BSc Environmental, Process & Energy Engineering, MCI (SS 2024, SS 2025)
- *Databases and Visualisation of Data*, BSc Biotechnology & Food Engineering, MCI (WS 2025)

Master's theses

- Felix Mittendrein, MSc Biotechnology (Ecobiotech), MCI

Bachelor's theses

- Ann Sophie Mossauer: *Einfluss von Vergaserkohle auf die Biogasproduktion anaerober Vergärungssysteme im kontinuierlich betriebenen Reaktor*, BSc Environmental, Process & Energy Engineering, MCI, 2024
- Jakob Germann: *Effekt von Aktivkohle auf die Biogasproduktion*, BSc Environmental, Process & Energy Engineering, MCI
- Philipp Theis: *Validierung und Anpassung einer HPLC-Methode zur Quantifizierung pharmazeutischer Rückstände in Klär- und Faulturmschlamm*, BSc Environmental, Process & Energy Engineering, MCI
- Felix Mittendrein: *Einfluss von medikamentenbeladener und ungeladener Bio- und Aktivkohle auf anaerobe Vergärung*, BSc Biotechnology & Food Engineering, MCI

Grants & Scholarships

2024 **Dissertation Funding – Energy Transition & Circular Economy (DISS 2024-02)**
Christian Doppler Research Association (CDG), Vienna, Austria
Competitive doctoral funding, granted after international peer review, for the project *Effect of gasification residue-based activated carbon on biogas production after application as 4th treatment stage in wastewater treatment plants.*

Peer-Reviewed Publications

2026 **Bioresource Technology Reports**
Margreiter, C., Klotz, J., Mullaymeri, A., Mossauer, A. S., Hofmann, A., Wagner, A. O. (2026). Beyond the first use: Cascading benefits of bio-based gasification chars recycling in anaerobic digestion. *Bioresource Technology Reports*, 103177. doi:10.1016/j.biteb.2026.103177

2026 **Journal of Environmental Management**
Margreiter, C., Mullaymeri, A., Klotz, J., Mittendrein, F., Theis, P., Hofmann, A., Wagner, A. O. (2026). Compound-specific effects of micropollutant-loaded gasification char on anaerobic digestion: implications for closed-loop wastewater treatment. *Journal of Environmental Management*, 415, 130645. doi:10.1016/j.jenvman.2026.130645

2026 **Frontiers in Environmental Engineering**

Margreiter, C., Hofmann, A., Wagner, A. O. (2026). A routine method for the semi-quantitative determination of porous carbonaceous material in wastewater matrices. *Frontiers in Environmental Engineering*, 5, 1719988. doi:10.3389/fenve.2026.1719988

2026 **Polymers**

Marx, J., **Margreiter, C.**, Hettich, V., Urban, C., Wagner, A. O., Prem, E. M., Pham, T., Spruck, M., Back, J. (2026). PES/PVP multi-channel mixed-matrix membranes with embedded activated carbon for co-removal of microorganisms and extracellular DNA from wastewater effluent. *Polymers*, 18(10), 1219. doi:10.3390/polym18101219

2025 **Chemical Engineering Research and Design**

Bosch, D., Back, J. O., Spruck, M., Nohel, L., Gurtner, D., **Margreiter, C.**, Hofmann, A., Bockreis, A. (2025). Addressing Europe's new hunger for sustainable activated carbon in wastewater treatment: Micropollutant removal with residual wood-based adsorbents using different activation strategies. *Chemical Engineering Research and Design*, 217, 108–120. doi:10.1016/j.cherd.2025.03.021

2024 **Heliyon**

Margreiter, C., Probst, M., Prem, E. M., Hofmann, A., Wagner, A. O. (2024). Gasification chars and activated carbon: Systematic physico-chemical characterization and effect on biogas production. *Heliyon*, 10(10), e31264. doi:10.1016/j.heliyon.2024.e31264

Manuscripts in Preparation

in preparation **Reactor study**

Margreiter, C. et al. In-situ dosing of bio-based gasification char: effects on compound-specific digestion kinetics and microbial community structure during anaerobic digestion of pharmaceuticals and PFAS.

Conference Contributions

03/2026 **Poster**

Österreichische Abfallwirtschaftstagung, Vienna Airport, Austria, 25–27 March 2026

Germann, J., Senfter, T., Kostner, S., **Margreiter, C.** Effekt von Aktivkohle auf die Biogasproduktion mit Klärschlamm.

09/2025 **Oral presentation**

ISAM 2025 – 13th International Symposium of Anaerobic Microbiology

Evaluating the impact of bio-based gasification chars and activated carbon on anaerobic digestion in wastewater treatment.

07/2025 **Poster**

19th Mini Symposium Verfahrenstechnik & 10th Partikelforum, MCI Innsbruck, Austria, 1–2 July 2025

Margreiter, C., Mittendrein, F., Hofmann, A., Wagner, A. O. Desorption of micropollutants from bio-based gasification chars in wastewater treatment. ISBN 978-3-200-10576-8.

09/2024 **Oral presentation**
ÖBIKA Biochar Day, Vienna, Austria
Vergaserkohle und Aktivkohle: Systematische physikalisch-chemische Charakterisierung und Einfluss auf die Biogasproduktion.

07/2024 **Poster**
Mini Symposium Verfahrenstechnik 2024, TU Graz, Austria
A method for quantifying porous carbonaceous material in different wastewater matrices.

05/2023 **Poster**
ISAM – International Symposium of Anaerobic Microbiology, Innsbruck, Austria
Effect of gasification residues and activated carbon on biogas production in anaerobic digestion systems.

Research Interests

- Anaerobic microbiology; methanogenesis and biogas production
- Circular economy and sustainable waste management: biochar, gasification residues and activated carbon in wastewater treatment
- Analytical chemistry: HPLC/GC, method development and optimisation

Skills

Languages	German (native) · English (B2) · Latinum (basic Latin certificate)
Statistics & data	R / RStudio, SPSS, Statistica, Python (basic), MATLAB (basic)
Bioinformatics	Metagenomics: processing and analysis of microbial community sequencing data
Software	LaTeX, Inkscape, ArcGIS (basic), Microsoft Office
Laboratory	HPLC, GC; biochemical methane potential (BMP) tests; batch and continuous anaerobic reactor operation; physico-chemical characterisation of chars and activated carbon; adsorption experiments with micropollutants; quantification of carbonaceous material in wastewater matrices

Innsbruck, 13 November 2026

Christian Margreiter